

TABLE 2
DIRECTION COMMAND

Component	Name	Function
Push button	Forward	Send Command '0' to Arduino for DC motor direction
Push button	Stop	Send Command '1' to Arduino for DC motor direction
Push button	Reverse	Send Command '2' to Arduino for DC motor direction
Status indication	N/A	'Status: Forward' – When forwarding 'Status: Reverse' – When reversing 'Status: Stop' – When stopping, the motor
Vertical slider	N/A	Speed control for DC motor, Send Command '3 through 255' to Arduino for DC motor speed control

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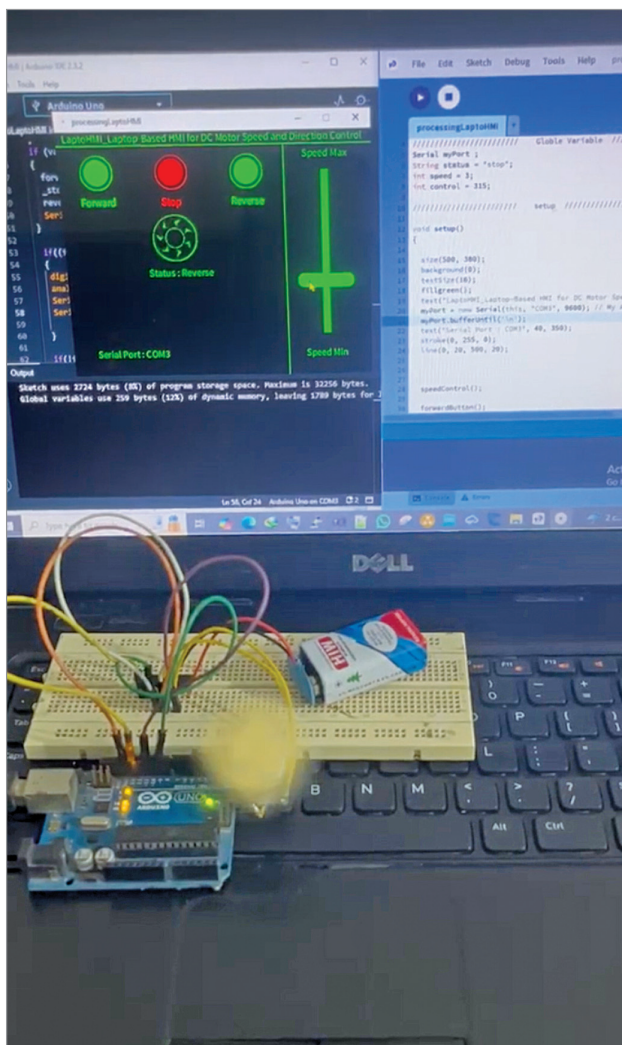


Fig. 6: Author's prototype connections and laptop HMI panel

three virtual push buttons, one status indicator, and one vertical slider. The push buttons are labelled as Forward, Stop, and Reverse. When the Forward button is clicked, Command '0' is sent from the

'Status: Forward' along with forward arrows inside a circle. When the motor is set to reverse, it displays 'Status: Reverse' with reverse arrows inside the circle. When set to stop state, a small

processing GUI to the Arduino via serial communication. The Arduino then relays this command to the L293D motor driver to set the DC motor's direction accordingly. Similarly, clicking the Stop button sends command '1', which stops the motor, while the Reverse button sends command '2' to change the motor's direction. If the Forward command sets the motor to rotate clockwise, then the Reverse command will cause it to rotate counterclockwise, or vice versa.

The status indicator functions as follows: When the DC motor is set to forward, the display shows

red rectangle appears inside the circle to indicate that the motor is halted.

The speed control bar is used to adjust the speed of the DC motor, allowing precise control of its movement in the selected direction. Refer to Table 2 for details.

Construction and testing

Connect the Arduino Uno to the laptop using a USB cable and identify the correct COM port in the device manager. Begin by opening the arduinoLaptopHMI.ino file in the Arduino IDE. Compile and upload the code to the Arduino Uno. Next, design the GUI layout using Processing. Once the layout is ready, run Processing and open the processingLaptopHMI.pde file. It is important to ensure that the COM port specified in the Processing code matches the one used by the Arduino in this setup; COM3 was used. Confirm that the Arduino is correctly programmed and connected before executing the Processing application. When the code runs successfully, a window appears as shown in Fig. 5. The GUI can now be used to control the motor's direction and speed (refer to Fig. 6). **EFY**

Pradipta Ghosh holds an M.E. in Electronics and Instrumentation. He currently serves as Section Engineer (Signal and Telecom) at Maha Mumbai Metro Operation Corporation Limited. He is passionate about electronics, MCU-based embedded systems, and IoT devices.

